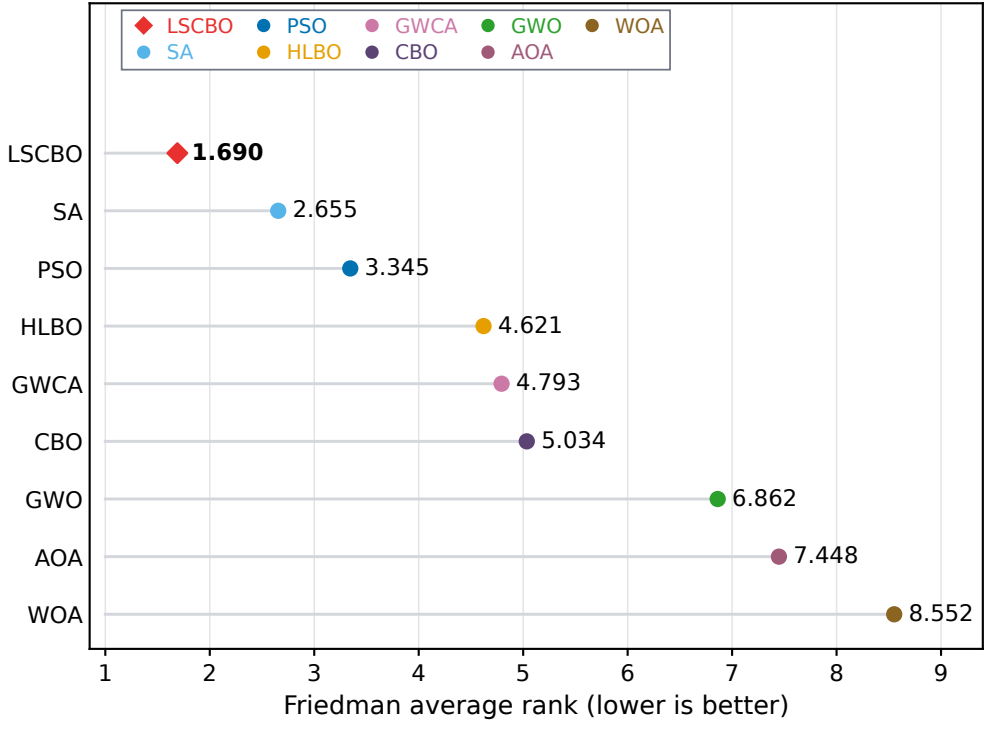
LSCBO
PSO
GWCA
GWO
WOA
SA
HLBO
CBO
AOA
LSCBO
SA
PSO
HLBO
GWCA
CBO
GWO
AOA
WOA
1.690
2.655
3.345
4.621
4.793
5.034
6.862
7.448
8.552
1
2
3
4
5
6
7
8
9
Friedman average rank (lower is better)